WORKED SOLUTION

STEP II 1993, Question 13

Mechanics • Simple Harmonic Motion

STEP II 1993, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q13

The key idea

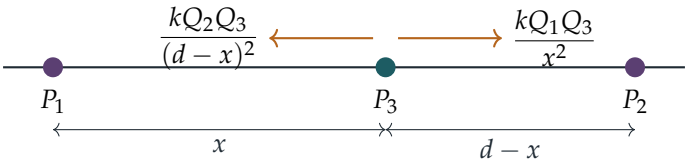
Put a coordinate on the moving bead and write down the net force on it: two inverse-square repulsions pulling in opposite directions. Equilibrium is where they balance. For the oscillation, displace the bead by a small y, use the given approximation on *both* forces, and the constant terms cancel by the equilibrium condition, leaving a restoring force proportional to y. That is exactly the equation of simple harmonic motion.

(The question says P_3 moves "between P_1 and P_3"; this is a slip in the paper for "between P_1 and P_2".)

Solution

Model. The beads are particles; the wire is smooth and horizontal, so the weight of P_3 and the normal reaction are both perpendicular to the wire and play no part in the motion along it. The only forces along the wire are the two electrical repulsions on P_3 (the fixed beads are held in place, so we need not worry about forces on them).

Measure x from P_1 towards P_2, so P_3 is at distance x from P_1 and $d - x$ from P_2, with $0 < x < d$.



P_1 pushes P_3 towards P_2 (positive direction) and P_2 pushes it back, so Newton's second law along the wire gives

$$m\ddot{x} = \frac{kQ_1Q_3}{x^2} - \frac{kQ_2Q_3}{(d-x)^2}. \qquad (*)$$

Equilibrium position

At equilibrium $x = a$ and the right-hand side of $(*)$ vanishes. Since k and Q_3 are positive we can divide by kQ_3:

$$\frac{Q_1}{a^2} = \frac{Q_2}{(d-a)^2} \quad \Longrightarrow \quad \left(\frac{d-a}{a}\right)^2 = \frac{Q_2}{Q_1}.$$

Both a and $d-a$ are positive (the bead is between the fixed beads), so we take the positive square root:

$$\frac{d-a}{a} = \frac{\sqrt{Q_2}}{\sqrt{Q_1}} \quad \Longrightarrow \quad \sqrt{Q_1}\,(d-a) = \sqrt{Q_2}\,a \quad \Longrightarrow \quad a = \frac{d\sqrt{Q_1}}{\sqrt{Q_1}+\sqrt{Q_2}},$$

as required. It will help to write $S = \sqrt{Q_1} + \sqrt{Q_2}$ and $b = d - a$, so that

$$a = \frac{d\sqrt{Q_1}}{S}, \qquad b = \frac{d\sqrt{Q_2}}{S}.$$

Notice the bead sits nearer the *weaker* charge, which makes sense.

Small oscillations

Let $x = a + y$ with y small, so $d - x = b - y$ and $\ddot{x} = \ddot{y}$. Using the given approximation (and the same approximation with y replaced by $-y$ and a by b),

$$\frac{1}{(a+y)^2} \approx \frac{1}{a^2} - \frac{2y}{a^3}, \qquad \frac{1}{(b-y)^2} \approx \frac{1}{b^2} + \frac{2y}{b^3}.$$

Substituting into $(*)$:

$$\begin{aligned} m\ddot{y} &\approx kQ_3\left[Q_1\left(\frac{1}{a^2} - \frac{2y}{a^3}\right) - Q_2\left(\frac{1}{b^2} + \frac{2y}{b^3}\right)\right] \\ &= kQ_3\left(\frac{Q_1}{a^2} - \frac{Q_2}{b^2}\right) - 2kQ_3\left(\frac{Q_1}{a^3} + \frac{Q_2}{b^3}\right)y. \end{aligned}$$

Ask yourself What happens to the terms that don't involve y?

They are exactly the equilibrium condition, so $\frac{Q_1}{a^2} - \frac{Q_2}{b^2} = 0$ and they cancel. This always happens when you expand about an equilibrium: the zeroth-order force is zero by definition.

Now simplify the coefficient of y using the expressions for a and b:

$$\frac{Q_1}{a^3} = \frac{Q_1 S^3}{d^3 Q_1^{3/2}} = \frac{S^3}{d^3\sqrt{Q_1}}, \qquad \frac{Q_2}{b^3} = \frac{S^3}{d^3\sqrt{Q_2}},$$

so

$$\frac{Q_1}{a^3} + \frac{Q_2}{b^3} = \frac{S^3}{d^3}\cdot\frac{\sqrt{Q_2}+\sqrt{Q_1}}{\sqrt{Q_1Q_2}} = \frac{S^4}{d^3\sqrt{Q_1Q_2}}.$$

Hence

$$\ddot{y} \approx -\omega^2 y, \qquad \omega^2 = \frac{2kQ_3S^4}{md^3\sqrt{Q_1Q_2}} > 0.$$

This is the simple harmonic motion equation, so for small displacements P_3 performs approximate SHM about $x = a$. (Released from rest at $y = y_0$, the motion is $y \approx y_0 \cos \omega t$.) The period is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{md^3\sqrt{Q_1Q_2}}{2kQ_3S^4}} = \frac{2\pi d}{S^2}\sqrt{\frac{md\sqrt{Q_1Q_2}}{2kQ_3}}$$
$$= \frac{\pi d}{S^2}\sqrt{\frac{4md\sqrt{Q_1Q_2}}{2kQ_3}} = \frac{\pi d}{(\sqrt{Q_1}+\sqrt{Q_2})^2}\sqrt{\frac{2md\sqrt{Q_1Q_2}}{kQ_3}},$$

which is the required result.

Checks

- **Symmetry:** with $Q_1 = Q_2$ we get $a = d/2$, the midpoint, as it must be. The period formula is symmetric in Q_1 and Q_2, which it should be too (swapping the charges just reflects the picture).
- **Dimensions:** kQQ'/r^2 is a force, so $kQ_3\sqrt{Q_1Q_2}$ has dimensions of force $\times$ length^2. Then $md\sqrt{Q_1Q_2}/(kQ_3) \cdot d^2/(\sqrt{Q_1}+\sqrt{Q_2})^4$ has dimensions mass $\times$ length/force $=$ time^2. Good.
- **Numbers:** with $Q_1 = 4$, $Q_2 = 1$, $d = 3$ and $k = Q_3 = m = 1$, we get $a = 2$ (check: $4/2^2 = 1/1^2$) and $T = 2\pi/\sqrt{3} \approx 3.628$. Integrating $(*)$ numerically from a tiny displacement, for many random choices of the constants, reproduces the period formula to better than one part in 10^4.

Common slips

- Expanding only one of the two forces: both depend on y, and both contribute to the restoring force (with the same sign).
- Getting the sign of the y term in $1/(b-y)^2$ wrong: replacing y by $-y$ in the given approximation gives $+2y/b^3$.
- Taking the negative square root in the equilibrium condition, which would put the bead outside the segment P_1P_2.

Beyond STEP

What we did is *linearisation*: near a stable equilibrium of a potential $V(x)$, the motion is approximately SHM with $\omega^2 = V''(a)/m$. Here $V = kQ_3\big(Q_1/x + Q_2/(d-x)\big)$, and you can check that $V''(a)$ gives the same ω^2. This idea, extended to several coordinates,

leads to normal modes of vibration, a central topic in first-year dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats small oscillations about equilibrium clearly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.